FORWARD DEPLOYED AI ENGINEERING

Capstone Project

Stage Four: Sprint Plan

Deciding who does what, in what order, and what finished means

A plan is not a list of tasks. It is a statement of sequence, ownership and completion that the team has actually agreed to.

WEEK TWO · DAY ONE

How to plan three weeks of work

You have already lost a third of the project to discovery and requirements, which is as it should be. What remains has to be sequenced carefully, because several parts of the build depend on others and the dependencies are not always obvious until you hit them.

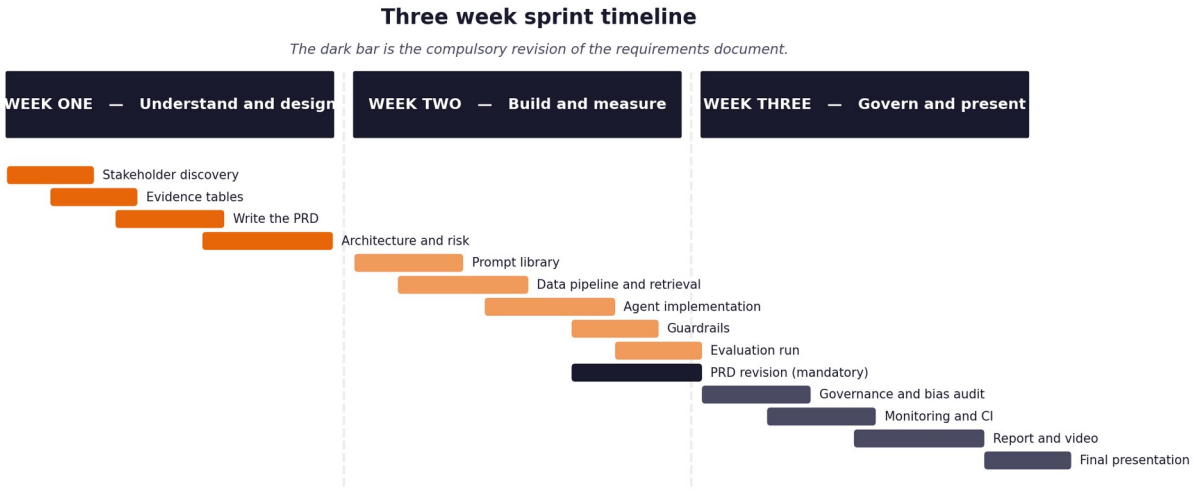


Figure 1. The shape of the three weeks, including the compulsory requirements revision.

Three rules that make the difference

1. Every item has an estimate in hours and a priority. When you run short of time, and you will, the priority column is what decides for you rather than whatever felt urgent that morning.
2. Every item has a definition of done written before work starts. Written afterwards, it describes whatever happened to be produced.
3. The retrieval layer and the evaluation harness come early, not late. Almost everything else depends on one or the other, and leaving the harness until the end means discovering your results too late to act on them.
4. The full unattended run over the hidden evaluation set happens in week two, not week three. It is the gate the project turns on, and it always exposes problems that never appear when you are testing one ticket at a time.

1 Section one: your capacity

Before you plan anything, work out honestly how much time you actually have. Plans built on optimistic capacity fail in week two rather than week three, which is worse, because by then you have committed to an architecture that assumed the time existed.

Week	Hours realistically available	Other commitments that week	Days you cannot work
Week one	Student input required	Student input required	Student input required
Week two	Student input required	Student input required	Student input required
Week three	Student input required	Student input required	Student input required

On working alone

Working individually removes the coordination overhead and replaces it with a different problem: nobody will notice when you have been stuck for six hours. Set yourself a rule that anything blocking you for more than an hour gets written down and either escalated or worked around, and check that list at the end of each day.

2 Section two: the backlog

List every item of work. Estimate in hours rather than in vague sizes, because hours can be compared against the capacity you recorded above. Mark dependencies explicitly, because the order matters more when there is only one of you.

ID	Item	Est. hours	Priority	Depends on	Definition of done
B-01	Environment and dependency setup	2	Must	—	Pinned setup works from clean checkout.
B-02	Data loading and normalization	3	Must	B-01	Load schema-compatible JSON inputs.
B-03	Discovery evidence synthesis	Student input required	Must	—	Five interviews and dataset analysis traceable.

B-04	PRD and requirement traceability	Student input required	Must	B-03	Requirements map to evidence and acceptance.
B-05	Intent/urgency classifier	6	Must	B-02	Separate predictions/confidences and metrics.
B-06	Documentation retrieval	5	Must	B-02	Top sources returned with IDs.
B-07	Routing/ confidence policy	4	Must	B-05,B-06	Uncertain/high-risk escalates.
B-08	Response generator integration	4	Should	B-06,B-07	Optional provider; failure is safe.
B-09	Guardrails and kill switch	6	Must	B-08	Guardrail block and kill-switch behavior.
B-10	SQLite decision logging	3	Must	B-07	One trace per ticket.
B-11	Evaluation harness	6	Must	B-05,B-10	Configurable paths; metrics and per-ticket output.
B-12	Automated tests	4	Must	B-11	Visible suite passes.
B-13	Visible validation evaluation	2	Must	B-12	One documented run; no tuning against results.
B-14	Governance and risk review	4	Must	B-09,B-13	Risk register and incident procedure.
B-15	Report/ workbook/	NEED	Must	All	Required files present;

	package				student confirms actuals.
B-16	Live video recording	NEED	Must	All	Student records 18–22 min video with 7+ min demo.
B-17	Hidden evaluation preparation	3	Must	B-11	Harness supports external input/output paths; hidden set not available.
B-18	CI/monitoring configuration	NEED	Should	B-12	Pack asks for CI and monitoring; production setup status disclose.
B-19	Human response review (>=50, two assessors)	NEED	Must for safety claim	B-08	Two independent ratings complete; outstanding.

3 Section three: the week two plan in detail

Day	What will be finished by the end of the day	Hours	Risk to this
Monday	Environment, data review, discovery	Student actuals required	Actual schedule unknown
Tuesday	PRD, evaluation plan, architecture	Student actuals required	Actual schedule unknown
Wednesday	Classification and retrieval	Student actuals required	Actual schedule unknown
Thursday	Routing, generator, guardrails	Student actuals required	Provider access/testing unknown
Friday	Harness, evaluation, revision	Student actuals required	Human review and hidden set unavailable

4 Section four: the week three plan in detail

Day	What will be finished by the end of the day	Hours	Risk to this
Monday	Governance, report and workbooks	Student actuals required	Personal records needed
Tuesday	Source cleanup and package	Student actuals required	Clean checkout should be verified
Wednesday	Record walkthrough/demo video	Student actuals required	Video not recorded yet
Thursday	Review final deliverables	Student actuals required	Resolve student-input fields
Friday	Submit package	Student actuals required	Deadline year not stated in pack

5 Section five: what you will drop if you run out of time

You will run out of time. Every student does. Deciding now what gets cut is far better than deciding in a panic on the final Wednesday, and it protects the parts that carry the most marks.

Item	Cut order	Consequence of cutting it	What you will say about it in the report
Live provider demo	1st to go	No live generated answer in demo	Provider integration exists but live path not verified.
Dashboard polish	2nd to go	Less visual monitoring	Decision logs/metrics remain available; state monitoring gap.
Additional prompt variants	3rd to go	Fewer comparisons	Versioned prompt library includes current design.
Human quality audit	Do not cut	Safety/citation claims remain unverified	Clearly state audit outstanding; do not claim target.
Effort log actuals	Do not fabricate	Workbook incomplete until student supplies records	Add actuals from personal records.

6 Section six: the daily check-in record

Ten minutes at the start of each day, answering the same three questions. Keeping this record is what makes your effort log accurate rather than reconstructed, and it is where you will notice a blocker that has quietly consumed two days.

Date	Finished since yesterday	Doing today	Blocked by
Student actual date required	Student to complete from actual records	Student to complete from actual records	Personal check-in history unavailable
Student actual date required	Student to complete from actual records	Student to complete from actual records	Personal check-in history unavailable
Student actual date required	Student to complete from actual records	Student to complete from actual records	Personal check-in history unavailable
Student actual date required	Student to complete from actual records	Student to complete from actual records	Personal check-in history unavailable
Student actual date	Student to complete	Student to complete	Personal check-in history

required	from actual records	from actual records	unavailable
Student actual date required	Student to complete from actual records	Student to complete from actual records	Personal check-in history unavailable
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